

EETS DevOps – Test 3 OpenBook Exam Reference

Condensed, printable reference aligned exactly to Test 3 topics.

1. Terraform & Infrastructure as Code

- 1 Terraform = Infrastructure as Code (IaC) tool by HashiCorp
- 2 Configuration language: HCL (HashiCorp Configuration Language)
- 3 Core commands: terraform init, terraform plan, terraform apply
- 4 main.tf defines providers, resources, variables, outputs
- 5 Providers define target cloud (e.g., provider "aws" {})
- 6 Variables allow parameterization using variable blocks
- 7 Outputs expose deployed resource info (e.g., web_public_url)
- 8 GitHub integration uses Personal Access Tokens (PAT)
- 9 AWS CodePipeline used to deploy Terraform labs

2. Docker & AWS Integration

- 1 Dockerfile defines how an image is built
- 2 `docker build -t` creates a tagged image
- 3 Amazon ECR stores Docker images in AWS
- 4 ECR login: `aws ecr get-login-password | docker login`
- 5 `docker push` uploads images to ECR
- 6 Image tagging identifies versions
- 7 ECS supports EC2 mode and Fargate mode
- 8 Registry acts as versioned artifact store in CI/CD

3. ECS & Fargate

- 1 ECS EC2 mode runs containers on managed EC2 instances
- 2 Fargate is serverless container execution (no EC2 management)

4. Kubernetes & Minikube

- 1 Minikube emulates a local Kubernetes cluster
- 2 minikube start launches local cluster
- 3 Kubernetes resources defined using YAML manifests
- 4 kubectl get pods lists running pods
- 5 NodePort service exposes pods externally
- 6 kubectl apply -f deployment.yaml deploys resources

5. CI/CD Pipelines

- 1 CI/CD flow: Commit → Build/Test → Deploy
- 2 AWS CodeBuild compiles and tests code
- 3 AWS CodePipeline orchestrates deployments
- 4 GitHub provides source code and triggers pipelines
- 5 GitHub Actions workflows defined in `.github/workflows/*.yml`
- 6 Terraform outputs like `web_public_url` used as pipeline outputs
- 7 Manual approval gates supported in CodePipeline

6. DevSecOps & Security

- 1 CIA Triad: Confidentiality, Integrity, Availability
- 2 STRIDE threat model: Spoofing, Tampering, Repudiation, Info Disclosure, DoS, Elevation of Privilege
- 3 Shift Left: introduce security early in SDLC
- 4 SAST analyzes source code
- 5 DAST analyzes running applications
- 6 RASP monitors applications at runtime
- 7 CVE = Common Vulnerabilities and Exposures
- 8 CVSS scores vulnerability severity
- 9 Availability countermeasures: redundancy, backups, monitoring
- 10 AWS CodeDeploy manages traffic to updated containers